

Proposing the RecursiVerse Overlay Application for the MetaVerse

Lorenzo Donatiello*
Department of Computer
Science and Engineering
University of Bologna

Gustavo Marfia†
Department of the Arts
University of Bologna

ABSTRACT

In a still uncertain future for the MetaVerse, we dare to envision one of its possible future developments, the RecursiVerse. The RecursiVerse is an overlay application that may be built upon the MetaVerse, amounting to symmetrical virtual-real space where a human being may rely on human digital twins which may move, operate and recursively replicate to collaboratively perform multiple tasks. The RecursiVerse may hence extend what will eventually be possible thanks to the MetaVerse, providing a service to a society that poses increasing cognitive and perceptual challenges due to growing work-life imbalances and increasing cognitive loads.

Index Terms: Human-centered computing—MetaVerse—Human Digital Twins—Non-fungible Tokens

1 INTRODUCTION AND BACKGROUND

After Palmer Luckey's introduction of the Oculus Rift and Mark Zuckerberg's announcement of the MetaVerse, the hype for the technologies that fall under the umbrella term eXtended Reality (XR) has incredibly grown. The expectations are many and come from completely different fields, ranging from entertainment to industry 4.0, from archaeology to precision surgery, just to name a few. Where the growth of an embodied Internet may lead may appear an educated guess at the time of writing, when thinking of the evolution of the Internet itself, for example.

Extending Zuckenberg's vision regarding what the MetaVerse may soon become, *an embodied Internet, where instead of just viewing content — you are in it* [1], we build upon a recent proposal which envisions what may come next in the reality-virtuality continuum, a symmetrical reality made of both virtual and augmented worlds with their inverses, comprising both real and virtual agents based on independent minds [8]. Assuming such vision may amount to a step in the evolution of the MetaVerse, we here move further ahead bringing into the picture the concept of Human Digital Twins (HDTs), as conceived in [7]. Recalling the words of the authors of [7], *human Digital Twin Computing aims to digitally express not only a person's outer self, such as physical characteristics but also that person's inner self such as personality and thoughts*.

Symmetrical realities, HDTs, and, as we will briefly discuss in this work, Non-Fungible Tokens (NFTs) amount to three of the key technologies which could support the creation of the RecursiVerse overlay, in essence, an ecosystem comprising human beings as well as their digital twins, where all actors, human and digital, may seamlessly and trustfully operate, interacting with both what is real and/or virtual [3, 7, 8].

In the following we proceed to set the stage for the RecursiVerse, first providing practical reasons which may motivate its creation (Section 2) and then an abstract scenario which explains how it may work (Section 3). We then summarize the key challenges that the

creation of such overlay poses in Section 4. We finally conclude with Section 5.

2 FROM VIRTUAL ASSISTANCE TO TWINS ASSISTANCE

Recent works show how a pervasive use of digital technologies may negatively affect work-life balance [6]. This amounts to an important problem, as being continuously subjected to high cognitive loads can at once increase stress and also reduce the general satisfaction and performance of a person.

One of the possible answers to such types of problems includes the use of digital technologies with the implementation and use of assistants capable of relieving cognitive pressure [2]. Such an approach falls within a general transformation which may widely involve digital technologies as life assistants performing many different tasks, from grocery shopping to dealing with bureaucratic procedures, from driving to managing a smart home.

The appearance of the MetaVerse and HDTs may bring a new perspective to what a virtual assistant may be: not anymore an exogenous agent that advises or performs obeying to someone else's approach or algorithm but a piece of digital software that is modeled according to the inner and outer self of its owner. The existence of the MetaVerse, and of a symmetric reality as introduced in [8], is crucial to such an ecosystem: every human being may be empowered with an arbitrary number of digital twins capable of equally processing and acting on real and digital surroundings and objects. In the following, we provide more detail of how this may work.

3 RECURSIVERSE TASKS

In [7] an HDT is defined as a *model that reproduces the individuality and characteristics of humans (for example, a personality/thinking model that models behavioral tendency, personality, and values or an ability model that models perception, knowledge, language ability, physical ability, etc.)*. While here embracing such definition, this is here expanded as within a RecursiVerse an HDT may deal with both virtual and real objects and settings. In essence, an HDT should implement a symmetrical cognition framework, i.e., the cognition as defined in [8]: within a symmetrical reality system, an HDT and a human being can communicate with each other by breaking the boundary separating the virtual from the physical world.

Imagine hence an ecosystem like the one abstractly represented in Figure 1. For the sake of simplicity, we here portray a situation where a human being (or an HDT) may deal with either a real or a virtual environment at once, omitting mixed situations which might deserve a different design approach. Following Figure 1, at the root of the RecursiVerse, we have a given *Human being*, who may operate in the immediate surrounding reality ($Reality_H$) or one of the available virtual environments ($Virtual Reality_H$). Based on given triggers, an arbitrary number of HDTs may stem from *Human being*, namely *Human Digital Twin*₀, ..., *Human Digital Twin*_n. Each of such HDTs may operate in real environments (not necessarily the one surrounding *Human being*) or in other virtual environments. We hence have that a generic instance *Human Digital Twin*_j may operate on the couple ($Reality_{HDT_j}$, $Virtual Reality_{HDT_j}$), for example. What happens in a generic ($Reality_{HDT_j}$) or ($Virtual Reality_{HDT_j}$) may however require taking other actions in other physical or digital spaces: this is when another arbitrary number of

*e-mail: lorenzo.donatiello@unibo.it

†e-mail: gustavo.marfia@unibo.it

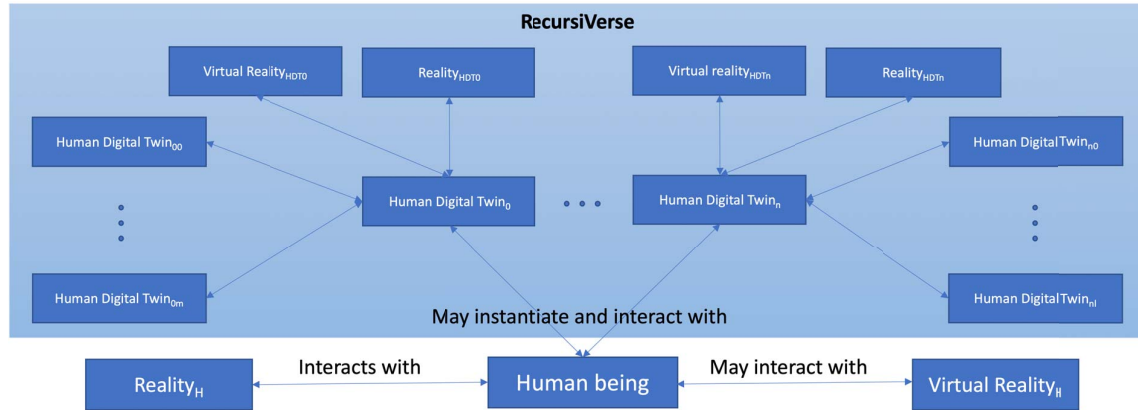


Figure 1: RecursiVerse recursion tree: a human being can either interact with and share information with a surrounding reality or virtual reality or trigger the RecursiVerse overlay, to power up her/his reach and extent.

twins may be instantiated, e.g., *Human Digital Twin_{j0}*, ..., *Human Digital Twin_{jm}*, to carry out such actions.

At an abstract level, the problems addressed by the approach portrayed in Figure 1 resemble the classical operations research ones of project and assignment planning. The approach itself, instead, based on a recursive instantiation of HDTs, points to a wealth of research that has appeared in the past decades on agent-based modeling [5]. The novelty of the RecursiVerse overlay amounts to the presence of human beings, not as external supervising actors, as designers or coders, but as actors that play on the same stage and at the same level as other software-powered ones. The question is whether a RecursiVerse may originate from existing research or whether, instead, new and unexplored paths should be followed to implement the depicted scenario. In the next Section we attempt to identify the key challenges and sketch a few of the enabling technologies which may contribute to giving an answer to such a question.

4 KEY CHALLENGES

Part of the key challenges posed by supporting the RecursiVerse overlay is discussed in [7, 8], these include the need of creating models expressing individuality and emotions such as tendencies regarding judgment and behavior, as well as supporting the exchange of multimedia information (e.g., visual images, audio, force, etc.) within the context of symmetrical reality systems.

We here concentrate on further key aspects which deserve particular attention: (a) shared identity, (b) trust and independence, (c) orchestration of distributed, synchronous and asynchronous actions, (d) sharing of information, and, (e) relationship between physical and virtual actors.

Regarding (a), we can summarize this point as follows: at the time of deepfakes, how may anyone be sure of the identity of the HDT that is operating on behalf of a given human being? The answer may be already available, NFTs are being used to enable digital art ownership on the blockchain [4]. Each token is unique, and that token proves ownership rights over a digital asset. Such a type of approach could be extended to HDTs, although it may leave several open questions. (b) certainly amounts to an open problem when considering that an HDT may be required to operate independently (e.g., spend money?), without a human being informed in real-time of the initiatives and the decisions that are taken by one of its HDTs. How may this probably desirable scenario be supported? To what degree may a human being trust its HDTs? This problem includes but also exceeds any technical issues, as it also poses legal and ethical ones. Point (c) follows (b), as the degree of trust and independence accorded to HDTs also imposes constraints on how many tasks and

actions may be orchestrated. Two particularly interesting aspects are also those introduced with (d), which includes the sharing and propagation of information among HDTs and towards their owner, and (e) which presents a problem that exceeds the credibility and trustworthiness of interactions that may occur among the actors, digital and virtual, involved in the RecursiVerse, and delves to also include their attractiveness, for example.

5 CONCLUSION

Many human beings are already relaying many core tasks to pieces of software that act on their behalf, from websites used for e-commerce to algorithms that now empower self-driving cars. As the MetaVerse rolls out, an interesting question amounts to whether the intertwining of real and virtual objects and actors will provide further means of supporting and assisting human beings' during their everyday lives. We here provide a vision that extends the one, already available, of digital assistants. With this work, we identify the key technologies which, within the MetaVerse, may serve the purpose of building the RecursiVerse overlay, a structure capable of indefinitely multiplying the capabilities of a single person and of funneling back results that may improve her/his wellbeing.

REFERENCES

- [1] Founder's letter 2021, <https://about.fb.com/news/2021/10/founders-letter/>, Accessed: 2022-01-03.
- [2] F. Brachten, F. Brünker, N. R. Frick, B. Ross, and S. Stieglitz. On the ability of virtual agents to decrease cognitive load: an experimental study. *Information Systems and e-Business Management*, 18(2):187–207, 2020.
- [3] S. Chevet. Blockchain technology and non-fungible tokens: Reshaping value chains in creative industries. *Available at SSRN 3212662*, 2018.
- [4] L. Kugler. Non-fungible tokens and the future of art. *Communications of the ACM*, 64(9):19–20, 2021.
- [5] C. M. Macal and M. J. North. Tutorial on agent-based modeling and simulation. In *Proceedings of the Winter Simulation Conference, 2005.*, pp. 14–pp. IEEE, 2005.
- [6] R. Palumbo. Let me go to the office! an investigation into the side effects of working from home on work-life balance. *International Journal of Public Sector Management*, 2020.
- [7] I. Toshima, S. Kobashikawa, H. Noto, T. Kurahashi, K. Hirota, and S. Ozawa. Challenges facing human digital twin computing and its future prospects. *NTT Technical Review*, 2020.
- [8] Z. Zhang. Symmetrical cognition between physical humans and virtual agents. In *2021 IEEE Conference on Virtual Reality and 3D User Interfaces Abstracts and Workshops (VRW)*, pp. 587–588. IEEE, 2021.